



Vidyavardhini's College of Engineering and Technology
Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VIII
Course Code:		Course Name:	RS

Name of Student:	BARI ANKIT VINOD
Roll No. :	61
Assignment No.:	2
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	5
Demonstrated Knowledge	3	2
Legibility	2	2
Total	10	9

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Mr. Ramesh Joshi

Signature :

Date :

Q. 1) Given dataset with parameters of class (Economy, Business), Sex (Male, Female) apply various pre-processing techniques based on their differences in collaborative filtering and explain each of them details.

→ we have : class $\rightarrow$ (Economy, Business) $\rightarrow$ Categorical
Sex $\rightarrow$ (Male, Female) $\rightarrow$ Categorical

$\therefore$ Both are nominal categorical variables.

Preprocessing for memory-based collaborative filtering
Memory based CF like User similarity using cosine or Pearson similarity works on iteration matrix.

Technique 1: Label Encoding
Convert categorical into integers
for class, Economy $\rightarrow 0$
 Business $\rightarrow 1$
for sex, male $\rightarrow 0$
 female $\rightarrow 1$

$\therefore$ Problem: label encoding introduces false ordinal meaning. CF similarity measures may assume, not recommended for similarity-based CF

Technique 2: One-hot encoding
Converts to binary vectors:
for class, Economy $\rightarrow 1$ & Business $\rightarrow 0$
for sex, male $\rightarrow 1$ & Female $\rightarrow 0$

$\therefore$ why good for CF? : No ordinal relationship, works well with cosine similarity, maintains neutrality. Also used in Hybrid CF, User similarity, boosting, demographic-based filtering.

- Technique 3 - demographic based similarity adjustment.
Instead of encoding into the rating matrix directly.

$$\text{Similarity} = \alpha (\text{Rating Similarity}) + \beta (\text{Demographic Similarity})$$

where,

: Same class $\rightarrow 1$ else 0

: Same Sex $\rightarrow 1$ else 0

This improves cold start users and sparse datasets.

- Technique 4 - Embedding encoding

Instead of one-hot (which increases dimensionality) convert -
class $\rightarrow$ embedding vector

sex $\rightarrow$ embedding vector

Eg. economics $\rightarrow [0.23, -0.11, 0.45, 0.01]$

business $\rightarrow [-0.55, 0.33, 0.90, -0.14]$

- Technique 5 - Feature scaling (If numerically encoded)

Sex-based preprocessing must consider bias -

- (i) Remove sensitive feature - Exclude sex to avoid biases recommended
- (ii) fairness regularization - add constraints in model -
minimize : Loss + λ (Bias Penalty)
- (iii) balance sampling - ensure equal male/female representation.

- Collaborative Filtering based on, users similar in behaviour will like similar items.

- Add side information
- Improve cold-start
- must be encoded carefully
- should not include artificial order
- must be evaluated for fairness.

2) Given is a collaborative filtering system that receives data in unsupervised fashion for sales growth with factors like weather, shipment vehicles and warehousing. Apply the clustering based mechanism specifically directed towards designing a recommendation system that handles the unsupervised patterns.

→ To design a clustering-based collaborative filtering system for sales growth using supervised data with factors

- weather
- shipment vehicles
- warehousing

we must structure the solution as an unsupervised recommendation architecture.

- The system receives data without labels, we need to -
 - detect hidden patterns
 - cluster similar sales conditions
 - recommend optimal logistics strategy
 - improve sales growth.

31. data preprocessing -

① weather data -

Temperature (numerical)

Rainfall (numerical)

Humidity (numerical)

weather type (Sunny, Rainy, Stormy)

- processing -

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

② shipment vehicles -

Truck

Air

Rail

Ship

If performance matrix exists - delivery time, fuel cost, delay frequency $\rightarrow$ normalize

• (c) warehousing -

features - capacity,

storage cost

Turnover rate

region

S3 - feature vector construction -

combine into single feature vector -

$$X = [\text{weather}, \text{vehicle}, \text{warehouse}]$$

$\therefore$ Now we have multidimensional contextual vectors.

S4 - apply clustering based mechanism -

(i) k-means clustering - group similar sales conditions together.

Algorithm -

$$\sum \|x_i - \mu_k\|^2$$

x_i = data point

μ_k = cluster centroid

(ii) Hierarchical clustering - Need tree based structure

Helps to detect micropatterns & macro patterns.

(iii) DBSCAN (density-based)

best when - irregular patterns

- noise in logistics data

- seasonal anomalies

• Converting clustering into recommendation -

$$\text{Sales Growth Rate} = \frac{\text{Current Sales} - \text{Previous Sales}}{\text{Previous Sales}}$$